

Lecture 33

NP, P vs NP

Complexity Class NP

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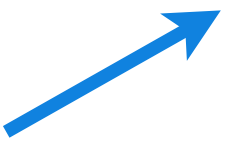
Defn: A decision problem L is in **NP** if $\forall x \in \{0,1\}^*$,

$$x \in L \implies \exists u \in \{0,1\}^{p(|x|)}, \text{ such that } A(x, u) = 1$$

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$x \in L \implies \exists u \in \{0,1\}^{p(|x|)}$, such that $A(x, u) = 1$  A on input (x, u) outputs 1.

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- Verifier cannot be fooled.

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Then, A on input $\langle x, u \rangle$, where $u \in \{0,1\}^{p(|x|)}$:

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Is $NP \subseteq P$? No one knows. It's a million dollar question!